

Reinforcement Learning

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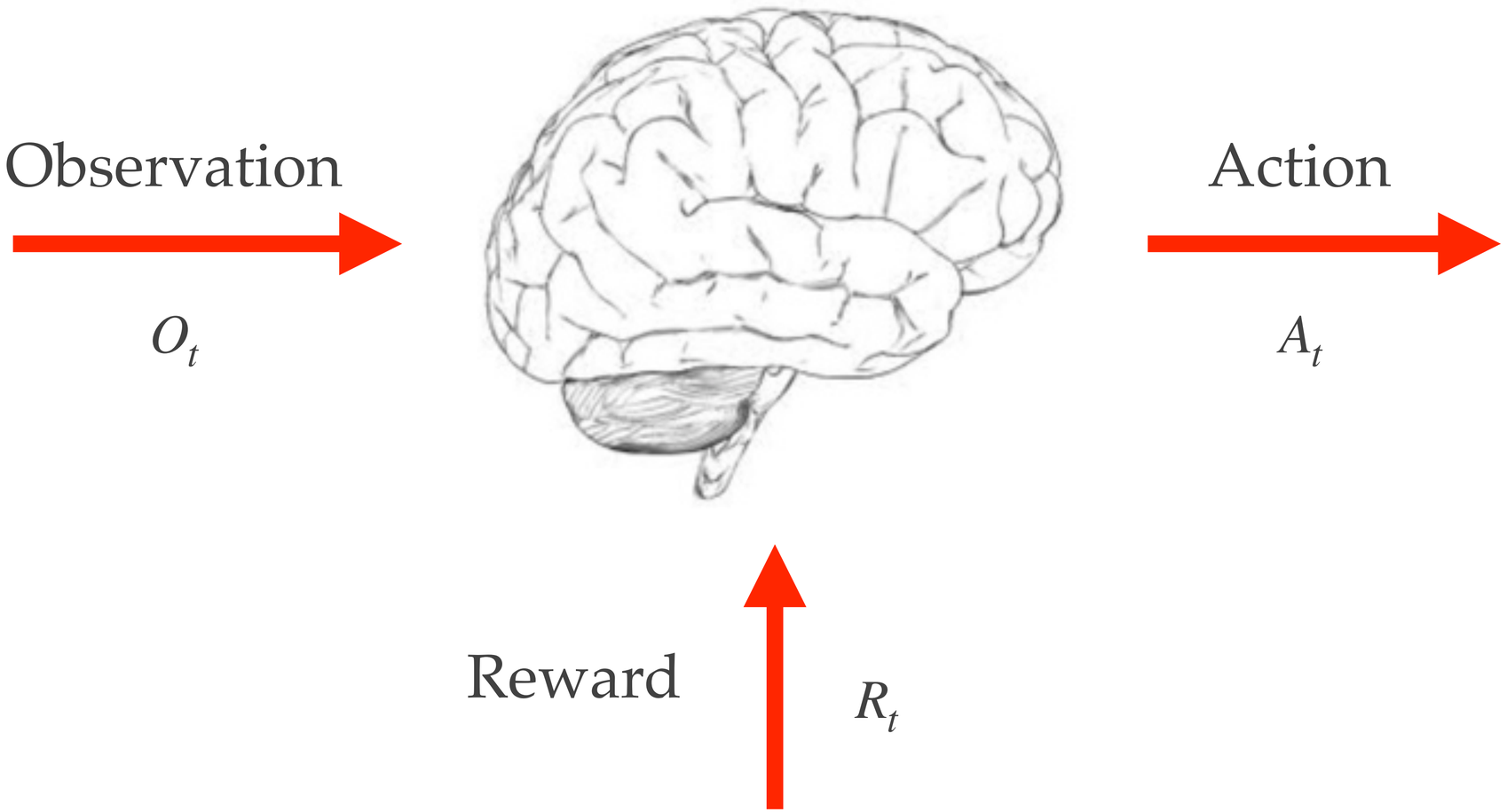
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Agenda

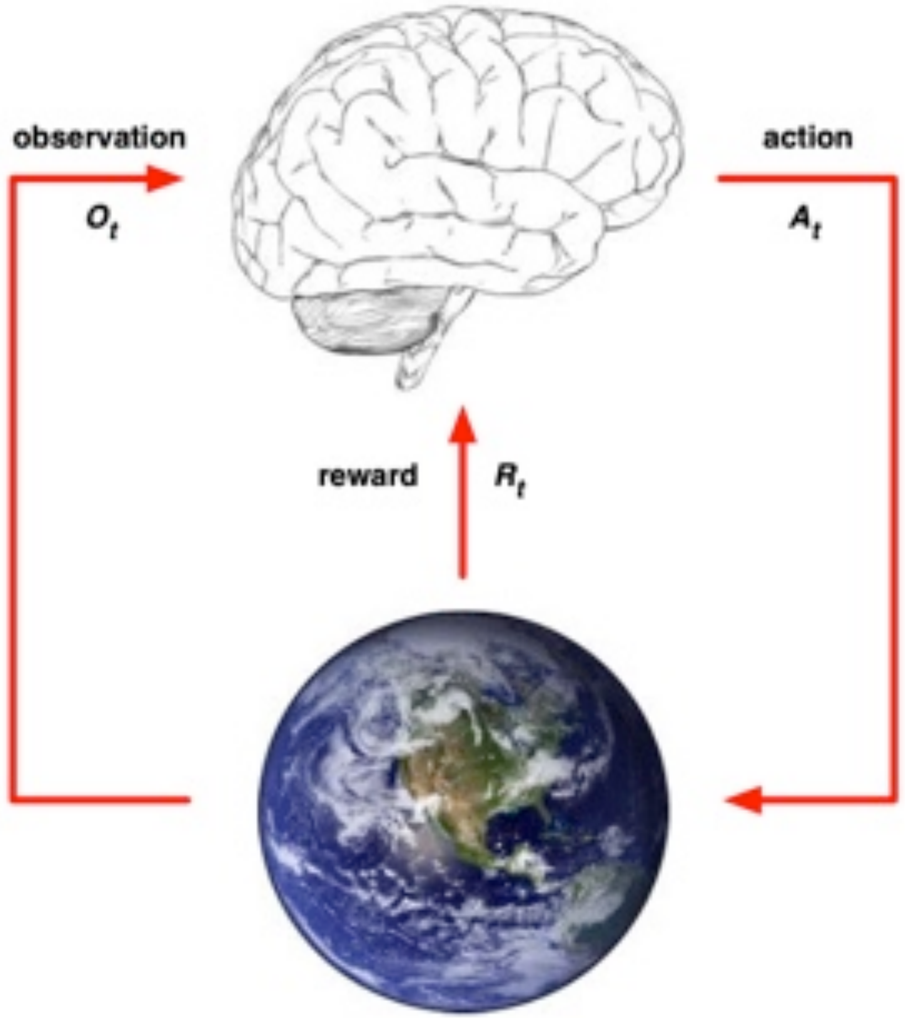
- Basic concepts
- Formulation

Basic Concepts

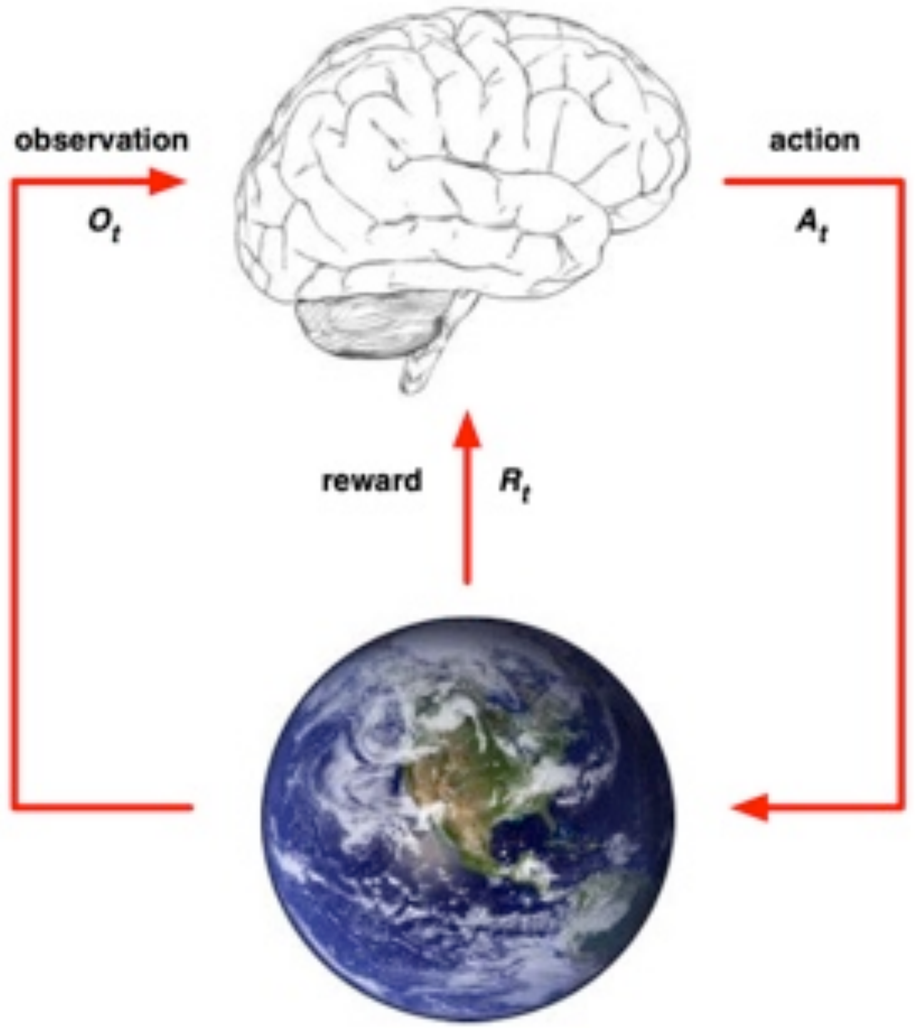
Agent



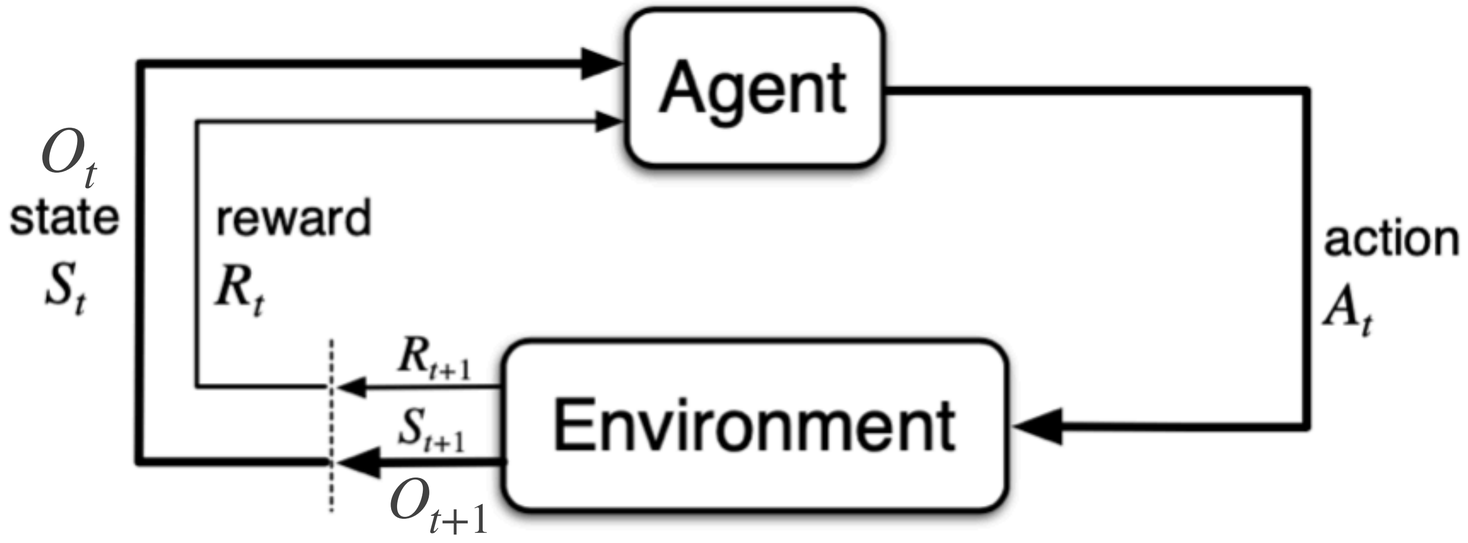
Agent & Environment



- The agent, at each step t,
  - Executes action A_t
  - Receives observation O_t
  - Receives reward R_t



- The environment
  - Receives action A_t
  - Emits observation O_{t+1}
  - Emits reward R_{t+1}



History

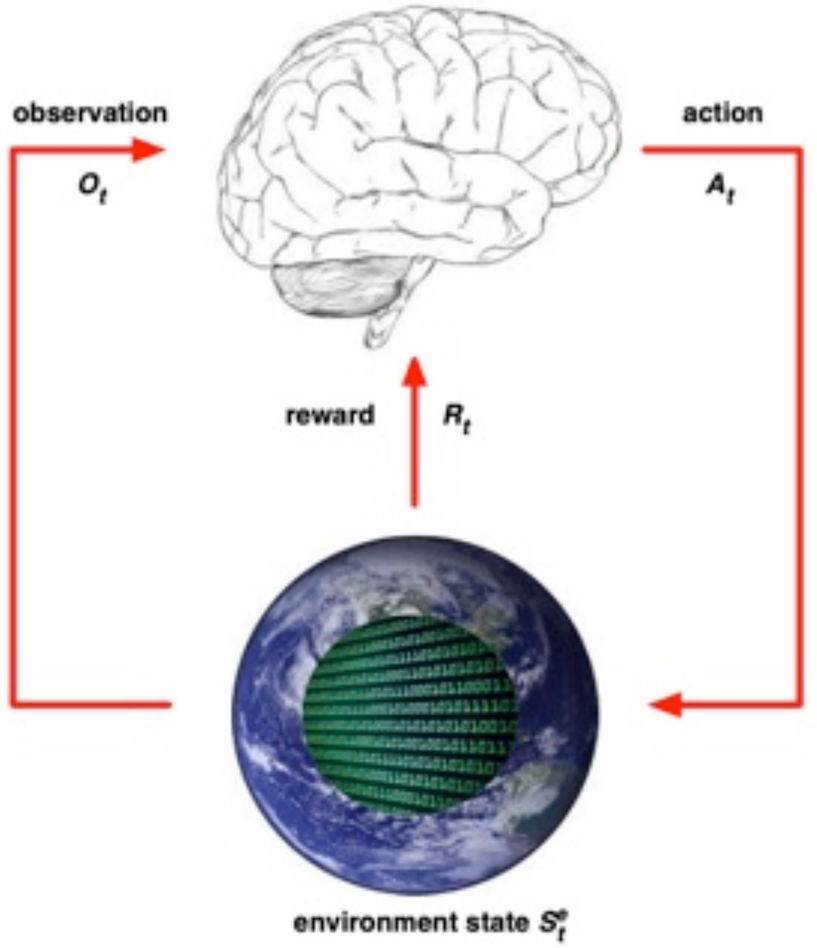
- History is the sequence of observations, actions, rewards
- $$H_t = O_1, R_1, A_1, \ldots, A_{t-1}, O_t, R_t$$
- i.e., everything that happened up to time t
- What happens next depends much on the history
  - The agent selects actions
  - The environment responds with observations/rewards

State

- State: the information used to determine what happens next
- Formally, state is a function of the history
- $$S_t = f(H_t)$$

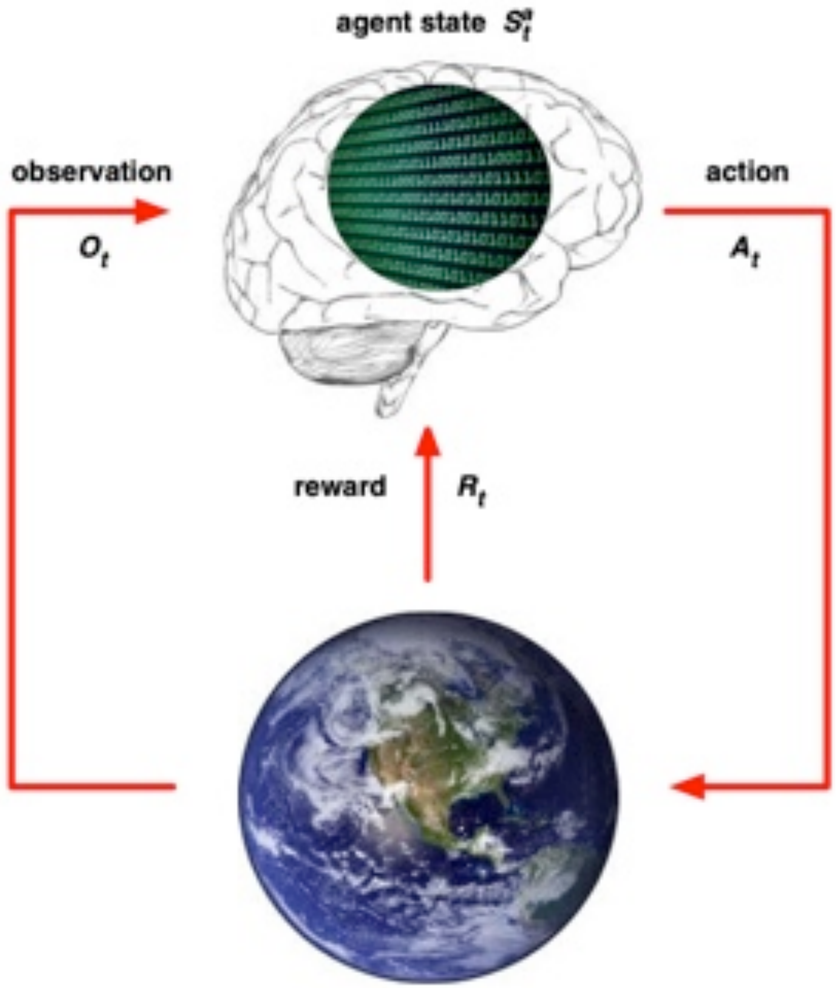
Environment State S_t^e

- Environment's private representation to emit next observation / reward
- Not visible to Agent
- Even if S_t^e is visible, some info may be irrelevant



Agent State S_t^a

- Agent's internal representation to pick next action
- Info used by RL algorithm
- Can be any function of history $S_t^a = f(H_t)$



Information State=Markov State

- An information state contains all useful information from the history.
- Definition: A state is Markov iff

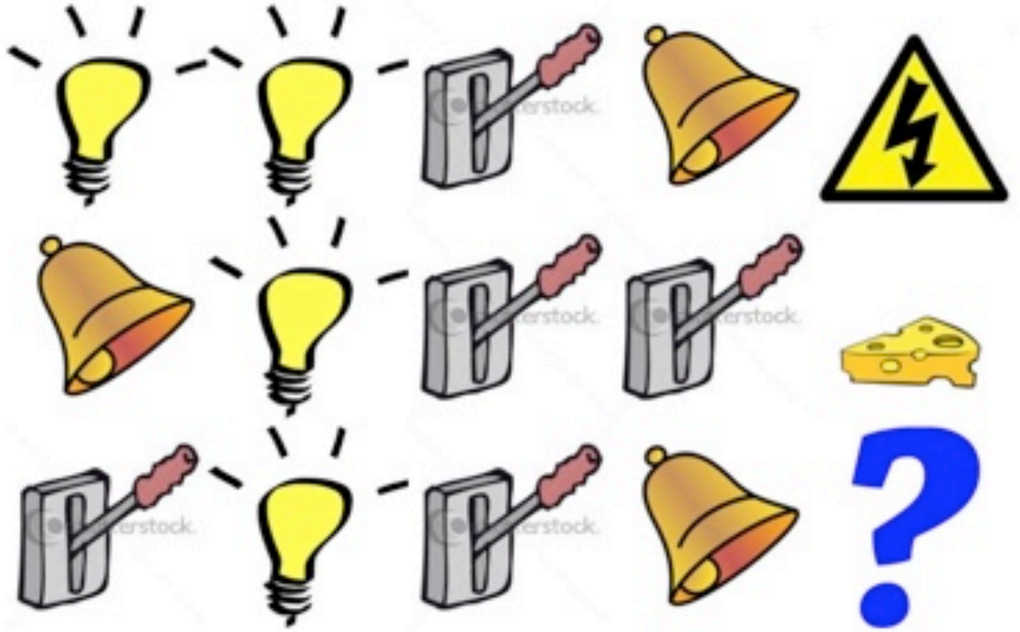
$$\mathbf{P}(S_{t+1} \mid S_t) = \mathbf{P}(S_{t+1} \mid S_1, \ldots, S_t)$$

- "The future is independent of the past given the present"
- $$H_{1:t} \rightarrow S_t \rightarrow H_{t+1,\infty}$$

- Once the state is known, the history becomes redundant
- The state is a sufficient statistic of the future

Choosing Information State: non-trivial





- What if agent state = last 3 items in sequence?
- What if agent state = counts for lights, bells and levers?
- What if agent state = complete sequence?

Markov Decision Process (MDP)

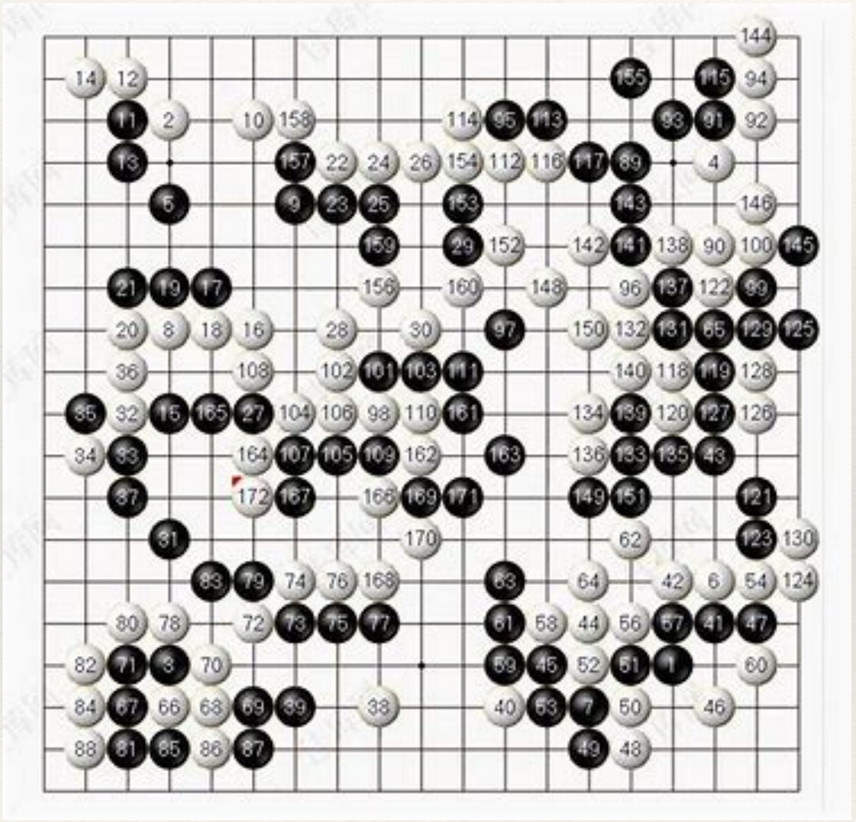
- MDP = fully observable environment (no hidden info)
- $$O_t = S_t^a = S_t^e$$
- Agent directly observes environment state
- Agent state = environment state = information state

Examples

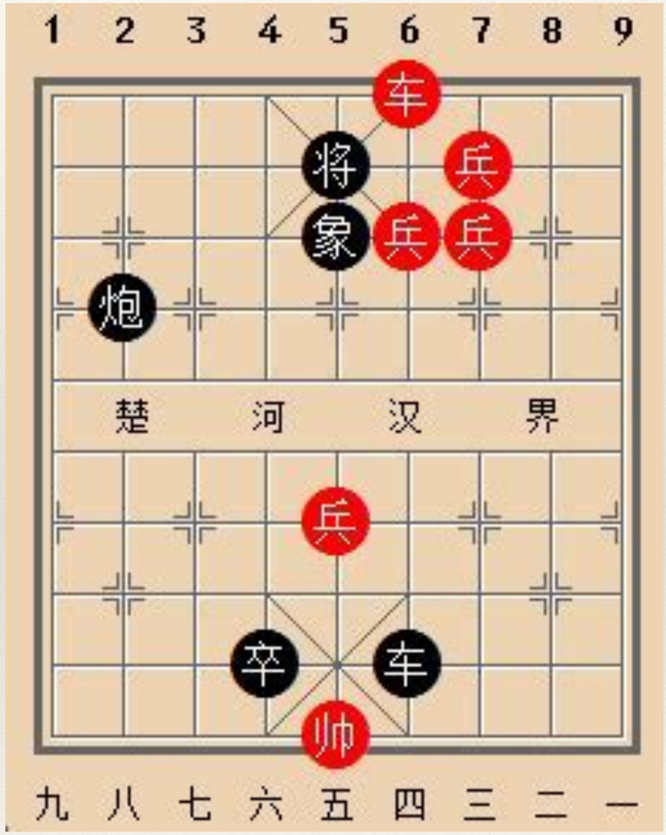
- The environment state S_t^e is Markov.
- (But invisible to Agent, so no use)
- The history H_t is Markov
- (But too big to handle in practice)

Need to choose carefully in practice

Examples of MDP







Markov?

- The current state is ALL you need to take actions

In practice

- How to choose S_t ?
- Current state, the history, or else?
- e.g. AlphaGo: the last 8 rounds

We focus mainly on MDP in this course

No MDP, No RL

Partially Observable Markov Decision Process (POMDP)

- POMDP = partially observable environment, i.e.,
- agent state $S_t^a \neq S_t^e$ (In fact, $S_t^a \subset S_t^e$)
  env state

Blackjack （21点）



Dealer’s card is not shown

Majoon （麻将）



Only observe public cards

Lost in the city



All streets look similar. No GPS.

How to get out of this?



All trees look the same

Stock price prediction



Alibaba

How about internal management, politics?

Agent must construct its own state representation S_t^a

- Complete history: $S_t^a = H_t$
- Beliefs of environment state:
- $$S_t^a = (\mathbf{P}[S_t^e = s^1], \ldots, \mathbf{P}[S_t^e = s^n])$$
- A recurrent neural network (RNN)
- $$S_t^a = \sigma(S_{t-1}^a W_s + O_t W_O)$$

Formulation

Data

- $S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \ldots\ldots$
  - $S_t \in \mathcal{S}$, state space
  - $A_t \in \mathcal{A}(s)$, action space
  - $R_t \in \mathcal{R}$, action space

Dynamics

which distribution

- $p(s', r \mid s, a) := \mathbf{P}[S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a)$

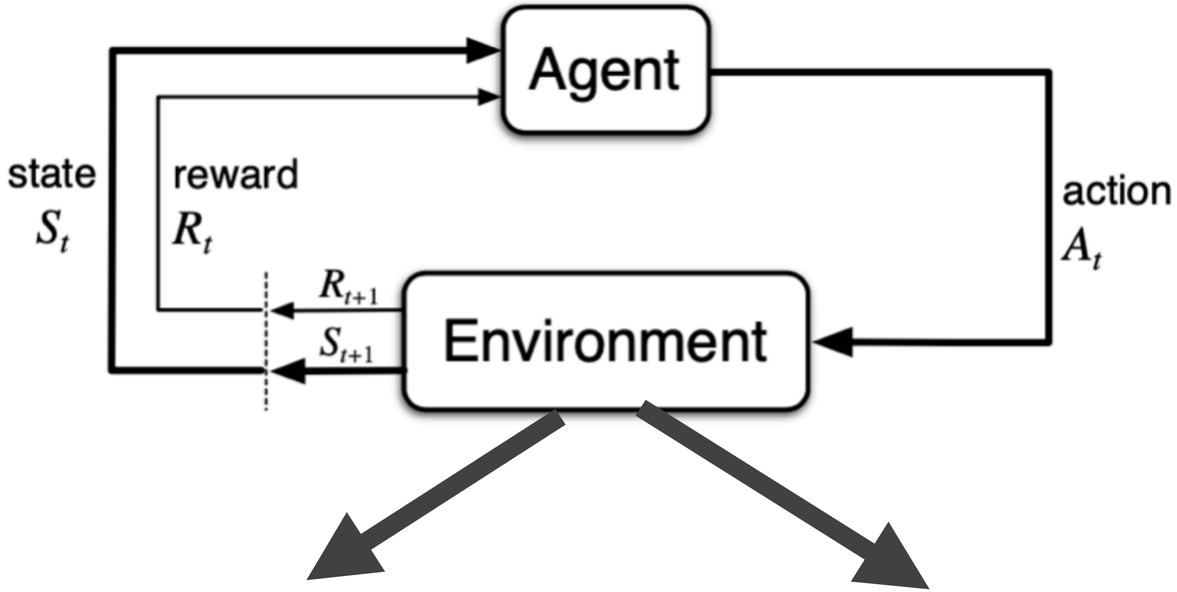
- for $s, s' \in \mathcal{S}$, $r \in \mathcal{R}$, $a \in \mathcal{A}(s)$.

Useful expressions

transfer probs to reward using expectation

- $p(s' \mid s, a) := \mathbf{P}[S_{t+1} = s' \mid S_t = s, A_t = a)$
- $p(r \mid s, a) := \mathbf{P}[R_{t+1} = r \mid S_t = s, A_t = a)$
- $r(s, a) := \mathbf{E}[R_{t+1} = r \mid S_t = s, A_t = a)$
- $r(s, a, s') := \mathbf{E}[R_{t+1} = r \mid S_t = s, A_t = a, S_{t+1} = s')$

Model-based vs Model free



Model-based

- Dynamics $p(s', r \mid s, a)$ is known
- Can plan ahead

Model-free

- Dynamics $p(s', r \mid s, a)$ is not known
- Learn directly

Major Components of an RL Agent

- Policy: agent's behaviour function
- Value function: how good is each state and/or action
- Model: agent's representation of the environment

Policy

- Deterministic: $a = \pi(s)$
- Stochastic: $\pi(a \mid s) = \mathbf{P}[A_t = a \mid S_t = s]$

Value function

- Prediction of future reward
- Used to evaluate how good/bad states are, and to select between actions, e.g.
- $v_\pi(s) = \mathbf{E}[R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots\ldots | S_t = s]$

Model

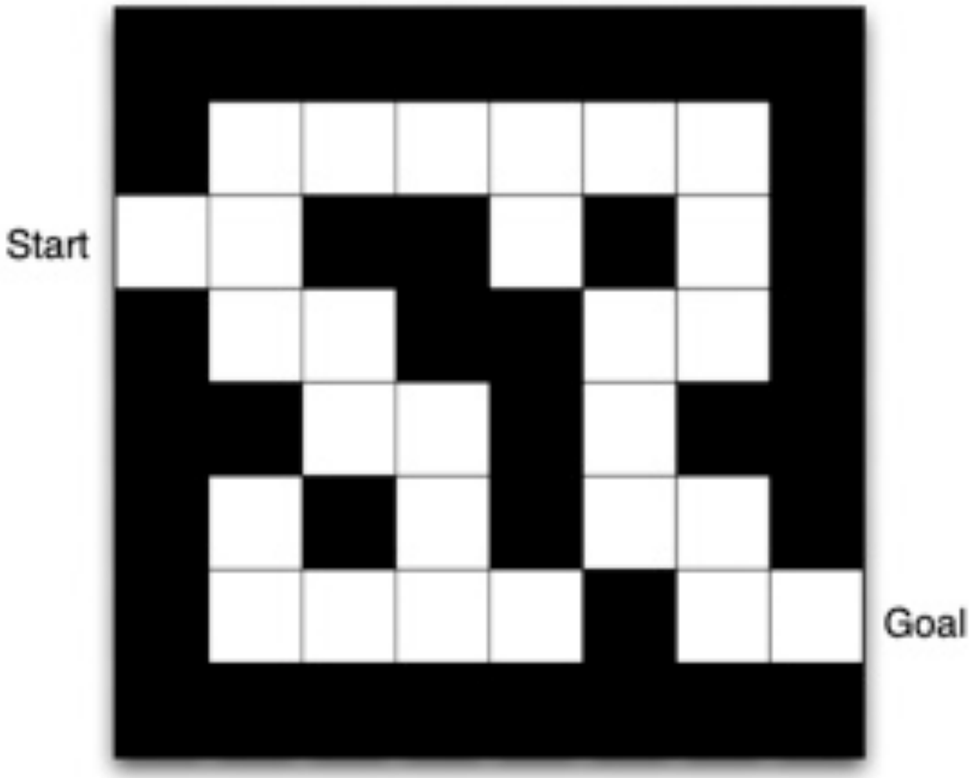
- Predicts what the environment will do next
- **P** predicts the next state

$$p(s' \mid s, a) := P^a_{ss'} = \mathbf{P}[S_{t+1} = s' \mid S_t = s, A_t = a]$$

- **R** predicts the next (immediate) reward

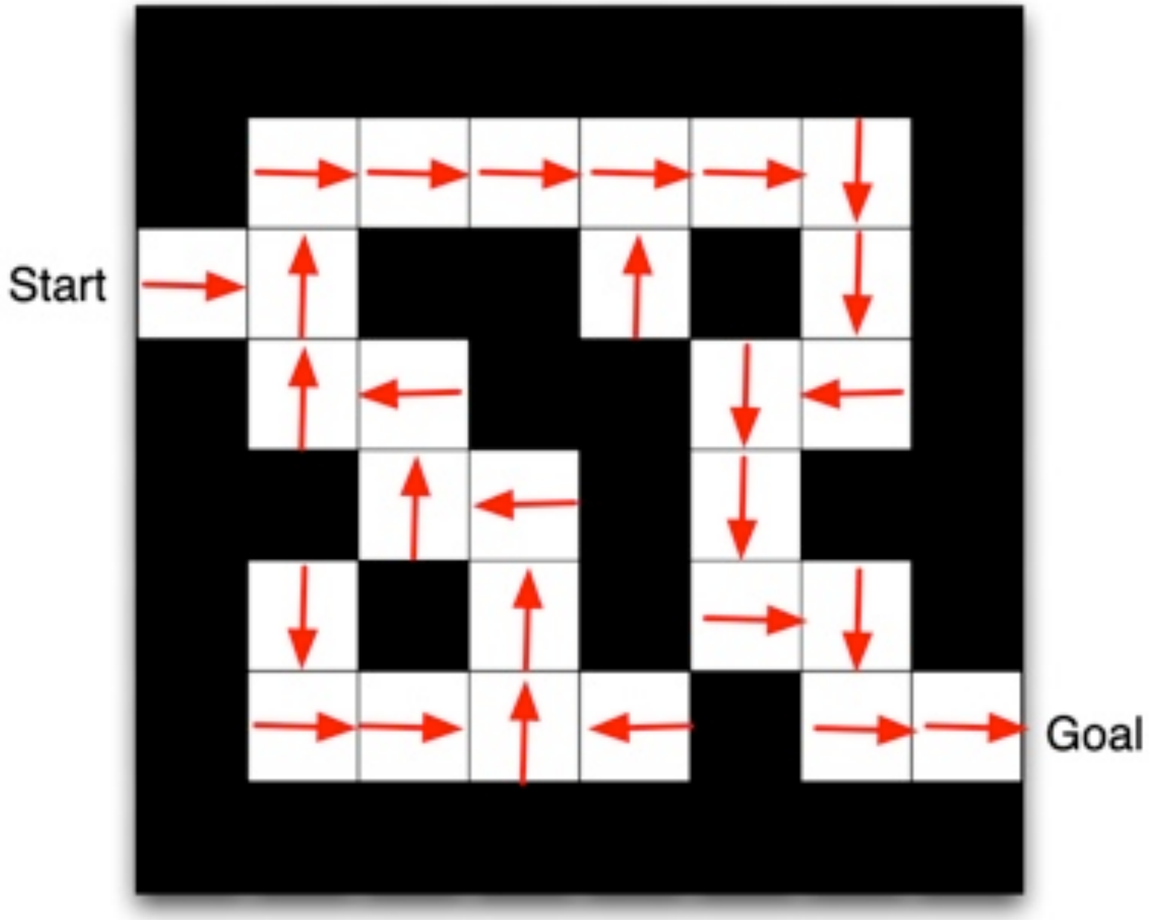
$$p(r \mid s, a) := \mathbf{P}[R_{t+1} \mid S_t = s, A_t = a]$$

Example: Maze



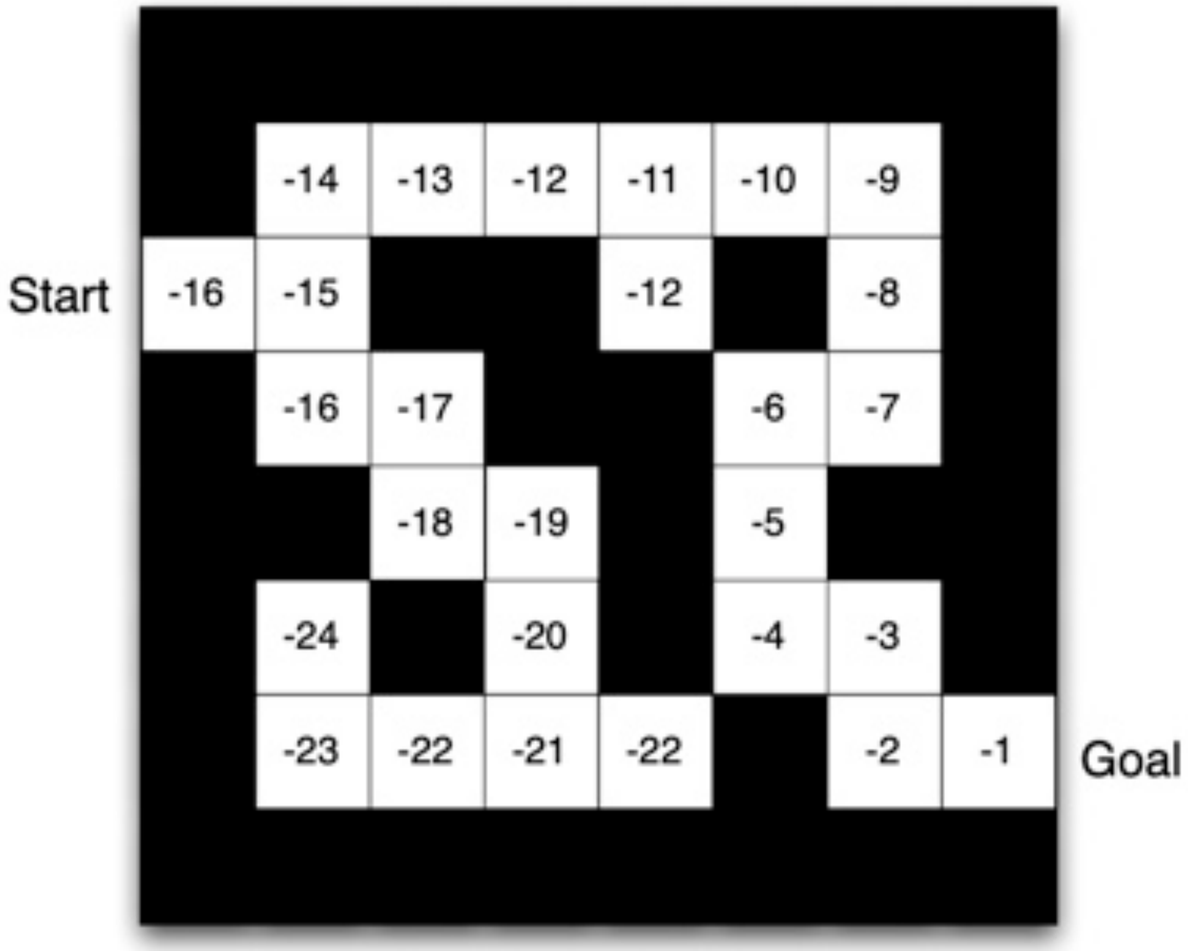
- Rewards: -1 per time-step
- Actions: N, E, S, W
- States: Agent's location

Maze: Policy



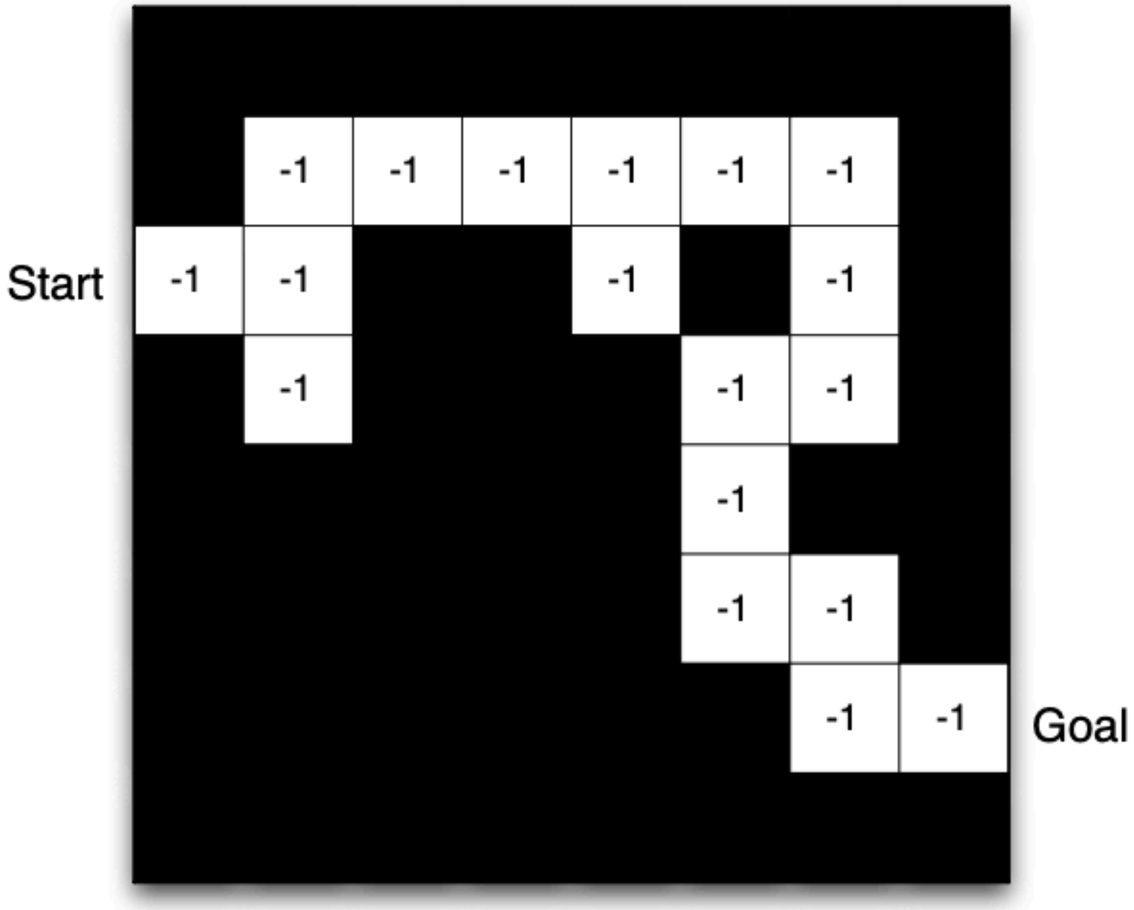
- Arrows represents policy $\pi(s)$ for each state s
- e.g. left: optimal policy π^*
- (Show how to find it later)

Maze: Value function



- Numbers represent value $v_\pi(s)$ of each state s
- (for the optimal policy π^* given in last page)

Maze: Model

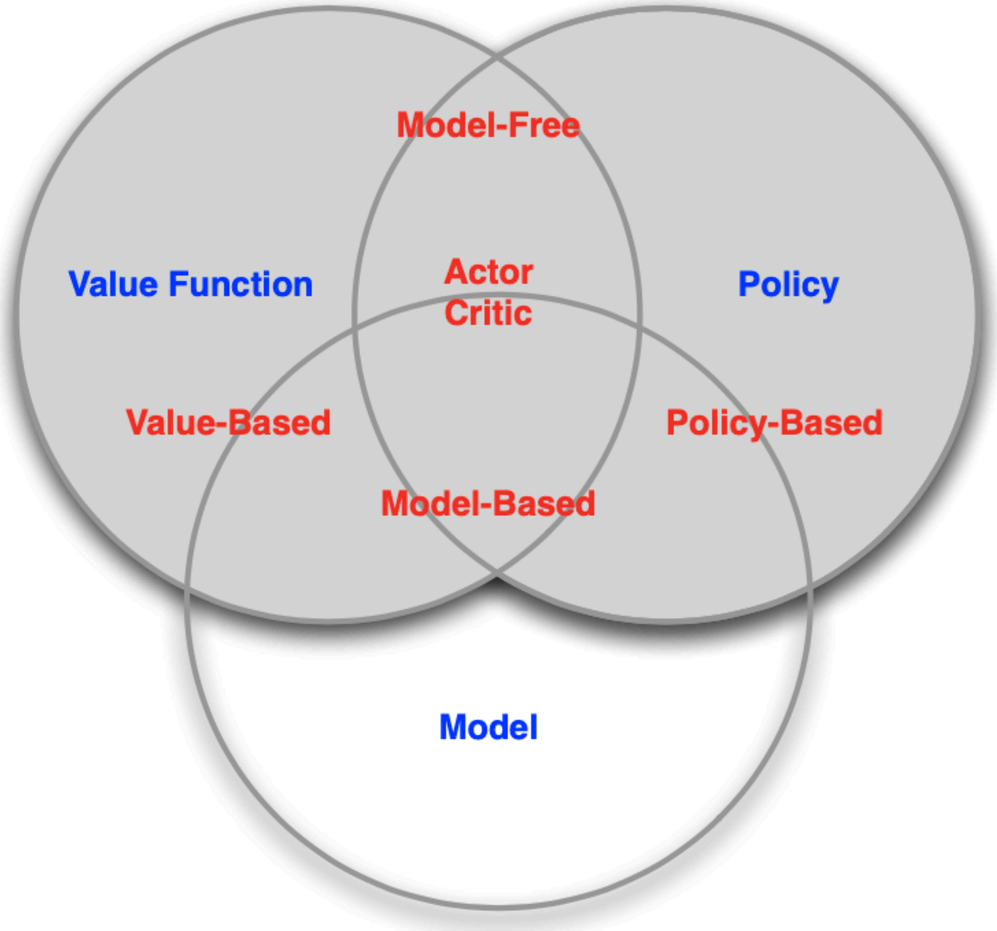


- Dynamics: how actions change the state
- Rewards: how much reward from each state

- Grid layout: transition model
- Numbers: immediate reward from each state

Categorizing RL agents

- Value-based
- Policy-based
- Actor-Critic



- Model-free
- Model-based

Learning & Planning

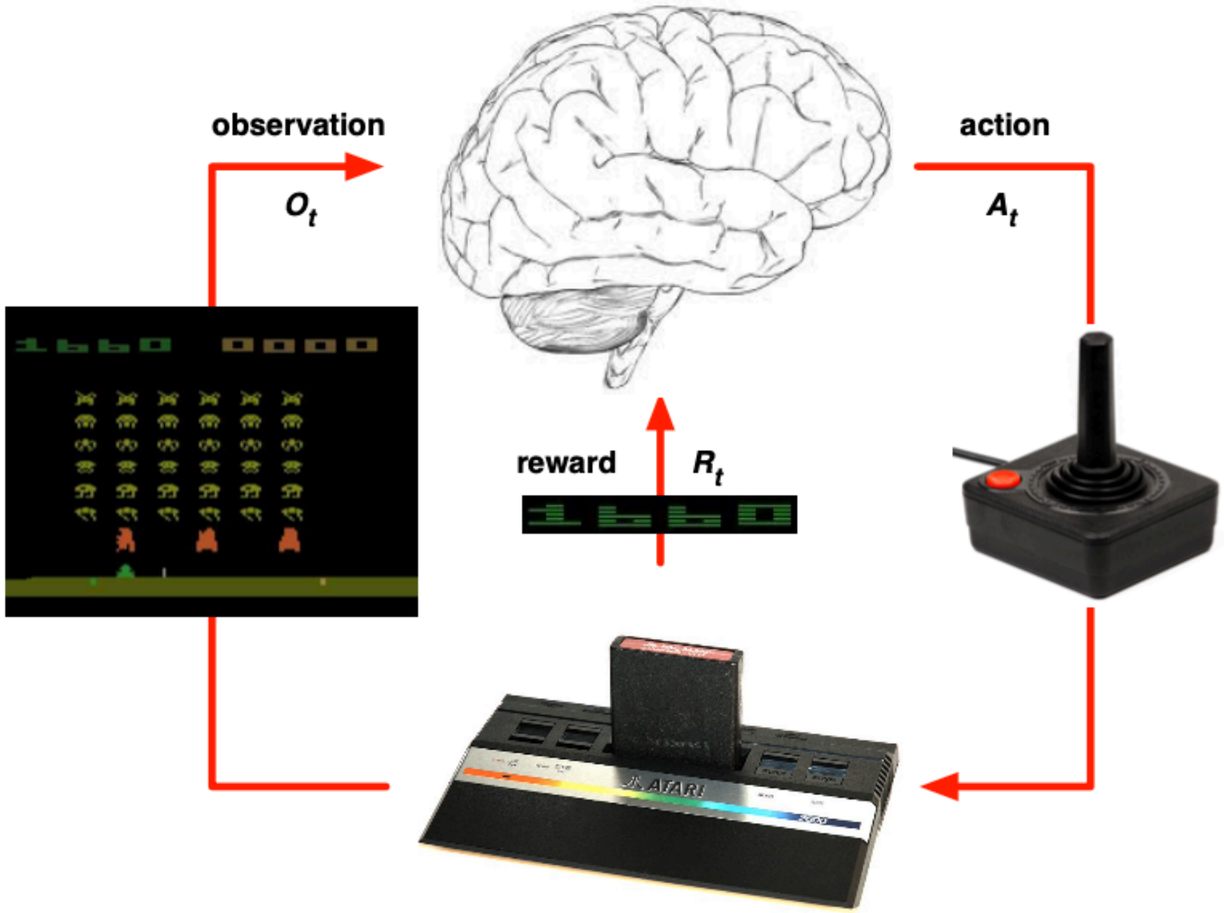
Reinforcement Learning

- Environment is initially unknown
- Agent interacts with the environment
- Agent improves its policy
- (Trial & error)

Planning

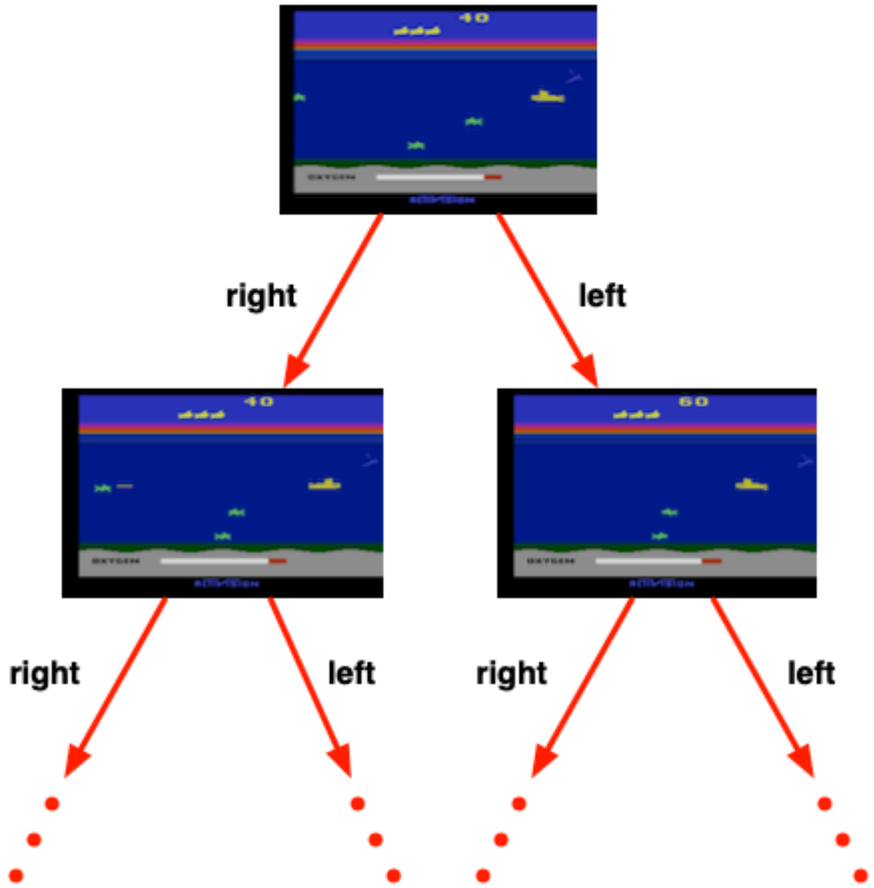
- Model of environment is known
- Agent performs computations with its model (without any external interaction)
- Agent improves its policy
- a.k.a. deliberation, reasoning, introspection, thought, search

Atari Example: Reinforcement Learning



- Rules of the game are unknown
- Learn directly from interactive game-play
- Pick actions on joystick, see pixels and scores

Atari Example: Planning

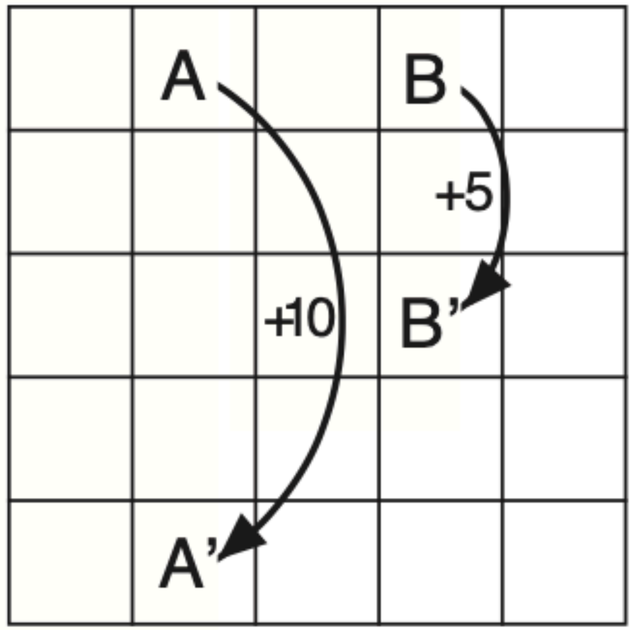


- Rules of the game are known
- Can query emulator e.g. if taking action a, what would the next state be? What would be the score?
- Plan ahead to find optimal policy

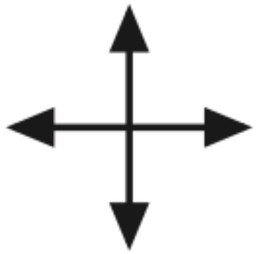
Prediction & Control

- Prediction: evaluate/estimate/predict the future (=expected reward) from following a given policy
- Control: find the best policy by optimising the future

Gridworld example: Prediction



(a)



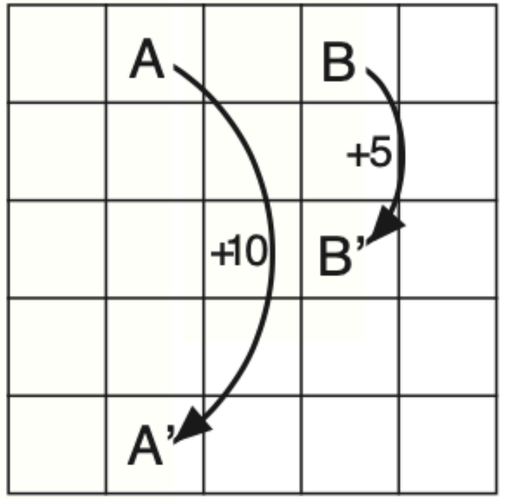
Actions

3.3	8.8	4.4	5.3	1.5
1.5	3.0	2.3	1.9	0.5
0.1	0.7	0.7	0.4	-0.4
-1.0	-0.4	-0.4	-0.6	-1.2
-1.9	-1.3	-1.2	-1.4	-2.0

(b)

What is the value function for the uniform random policy?

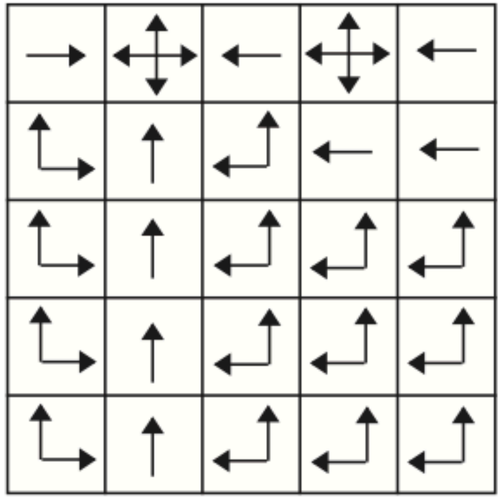
Gridworld example: Control



a) gridworld

22.0	24.4	22.0	19.4	17.5
19.8	22.0	19.8	17.8	16.0
17.8	19.8	17.8	16.0	14.4
16.0	17.8	16.0	14.4	13.0
14.4	16.0	14.4	13.0	11.7

b) v_*



c) π_*

What is the optimal value function over all possible policies?
What is the optimal policy?

Course outline

- Markov decision processes & planning
- Model-free policy evaluation
- Model-free control
- Reinforcement learning with function approximation & Deep RL
- Policy Search
- Exploration
- Advanced Topics

